

Practical exercises

— constructions

with compasses

and a straightedge

A Cambridge insert: the four constructions every later proof depends on.

LESSON 55

1 × 40 MIN

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S8 5.3

LESSON CLOCK

8'

14'

12'

6'

The tools	8 min
The four constructions	14 min
Building a triangle	12 min
Homework	6 min

LEARNING OBJECTIVES

- Copy a segment and an angle with compasses.
- Construct the perpendicular bisector of a segment.
- Construct the bisector of an angle.
- Construct a triangle from three given sides.

TEXTBOOK

National	pp. 149–151
Cambridge	Stage 8, pp. 53–56
Workbook	Exercise 5.3

01

Explanation

The tools

A **construction** uses only two tools: compasses to draw circles and to carry lengths, and a straightedge to draw lines. The straightedge is **not** a ruler — no measuring is allowed.

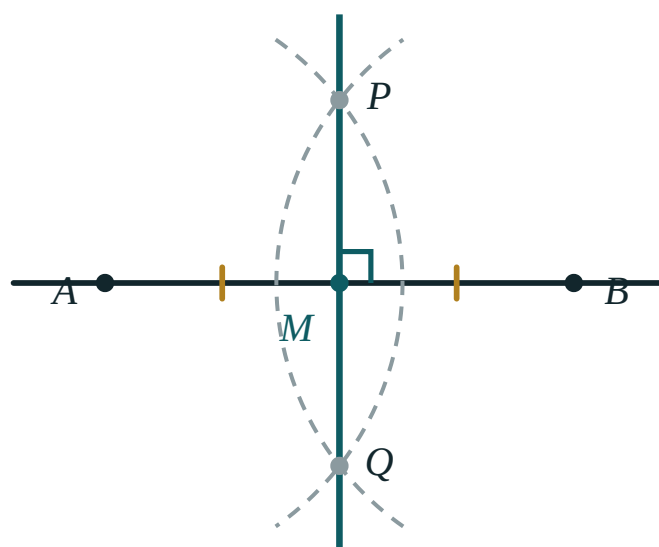
TOOL	MAY BE USED TO	MAY NOT
compasses	draw an arc, carry a length	measure in centimetres
straightedge	join two points	measure

LEAVE EVERY ARC VISIBLE

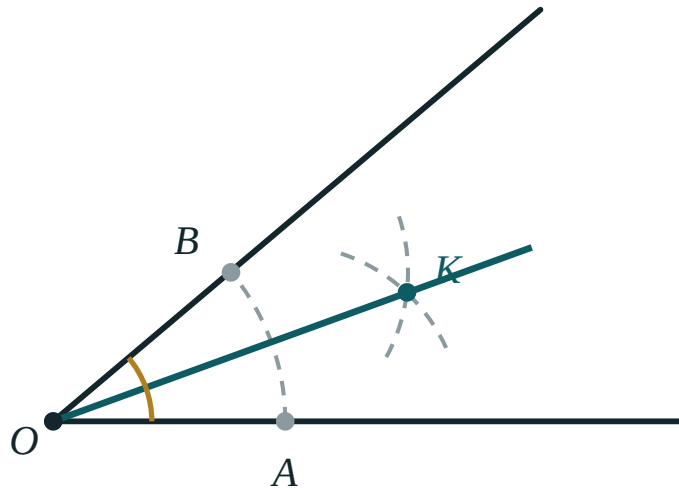
The arcs are the working. A finished construction with the arcs rubbed out cannot be marked, and scores less than an untidy one that shows them.

The four constructions

CONSTRUCTION	METHOD
copy a segment	open the compasses to it and step it off on a ray
copy an angle	one arc across both arms of the original and of the copy, then transfer the chord
perpendicular bisector of AB	equal arcs from A and from B , both sides; join the crossings
bisector of an angle	one arc across both arms, then equal arcs from the two points; join to the vertex



Equal arcs from both ends give the perpendicular bisector.



One arc across the arms, then two equal arcs, gives the bisector.

ALL FOUR ARE THE SAME IDEA

Two points equidistant from the same pair of points determine a line of symmetry. Every one of the four constructions is that statement applied somewhere.

Building a triangle

Given three sides a , b , c :

1. draw a segment $BC = a$;
2. from B , an arc of radius c ;
3. from C , an arc of radius b ;
4. the crossing is A ; join AB and AC .

By the third criterion, every triangle built this way from the same three lengths is congruent to every other — so the construction gives **the** triangle, not **a** triangle.

THE ARCS MUST ACTUALLY CROSS

If the two short sides together are less than the long one, the arcs miss and no triangle exists. The construction fails exactly when the triangle inequality fails.

02 Worked examples

EXAMPLE 1 Describe the construction of the perpendicular bisector of a segment AB .

- 1 Open the compasses to more than half of AB .
- 2 Draw arcs from A above and below the segment.
- 3 Draw the same arcs from B .
Same radius.
- 4 Join the two crossings — that line is the perpendicular bisector.

ANSWER

Equal arcs from both ends, joined

EXAMPLE 2 Construct a triangle with sides 4 cm , 5 cm and 6 cm .

- 1 Draw $BC = 6\text{ cm}$.
The longest side is easiest.
- 2 Arc of 4 cm from B .
- 3 Arc of 5 cm from C .
- 4 Join the crossing to both ends.
The arcs cross since $4 + 5 > 6$.

ANSWER

A unique triangle

EXAMPLE 3 Why does no triangle exist with sides 3, 4, 9?

- ① Draw the 9 *cm* side.
- ② Arcs of 3 and 4 from its ends.
- ③ $3 + 4 = 7 < 9$ — the arcs fall short.
- ④ They never meet, so there is no third vertex.

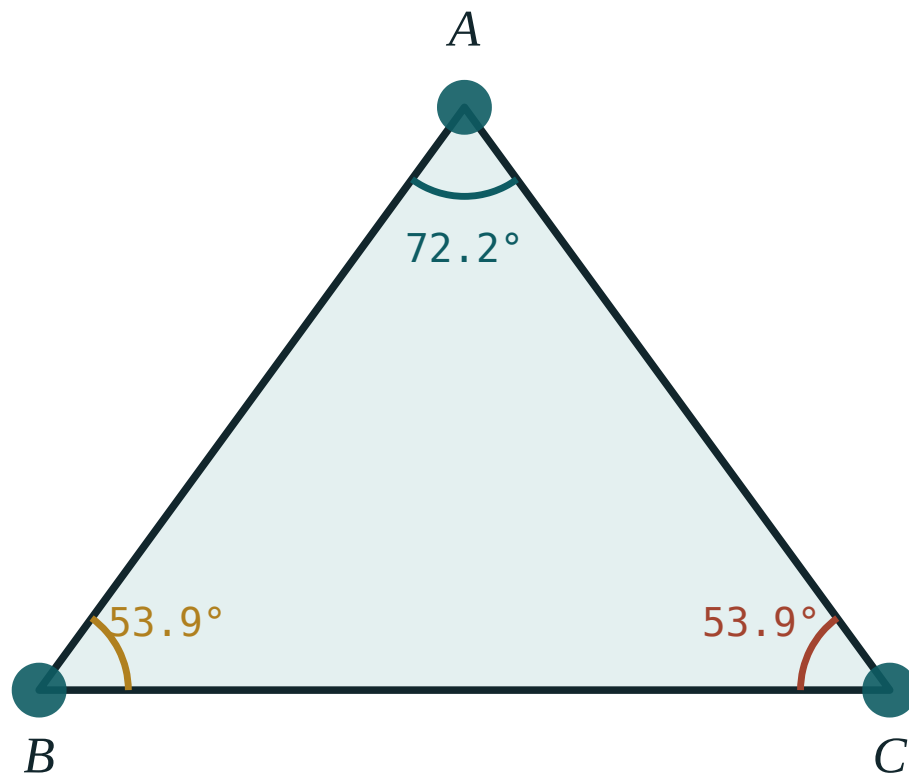
ANSWER

The arcs do not cross

03 Interactive model

Have the class construct the same triangle from the same three lengths and then lay the sheets on top of one another; every triangle coincides, and the third criterion is confirmed.

Building from three sides

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Compasses	Sirkul	Циркуль
Straightedge	Chizg'ich	Линейка
Arc	Yoy	Дуга
To copy	Ko'chirish	Перенести
Perpendicular bisector	O'rta perpendikulyar	Серединный перпендикуляр
Bisector	Bissektrisa	Биссектриса
Equidistant	Teng uzoqlikda	Равноудалённый

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A construction uses:

a ruler and a protractor

compasses and a straightedge

a calculator

a set square only

Construction arcs should be:

rubbed out

left visible

drawn in pen

omitted

The perpendicular bisector is found by:

measuring the midpoint

equal arcs from both ends

a protractor

folding only

A triangle from three sides is:

not always possible

always possible

possible in two ways

never unique

Sides 3, 4, 9:

one triangle

two triangles

no triangle

infinitely many

All triangles built from the same three sides are:

similar only

congruent

different

right-angled

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Tools allowed

ANSWER Compasses and a straightedge

2. May the straightedge measure?

ANSWER No

3. Should arcs be left?

ANSWER Yes

4. Sides 4, 5, 6: constructible?

ANSWER Yes

5. Sides 3, 4, 9: constructible?

ANSWER No

6. Sides 5, 5, 5: what triangle?

ANSWER Equilateral

7. Sides 3, 4, 5: what triangle?

ANSWER Right-angled

Medium · the standard question

7 PROBLEMS

1. Which construction gives the midpoint?

ANSWER The perpendicular bisector

2. Which gives two equal angles?

ANSWER The angle bisector

3. How many arcs for a perpendicular bisector?

ANSWER Four — two from each end

4. Construct 60° : method

ANSWER An equilateral triangle

5. Construct 30° : method

ANSWER Bisect 60°

6. Construct 90° : method

ANSWER A perpendicular

7. Construct 45° : method

ANSWER Bisect 90°

Hard · stretch and reason

7 PROBLEMS

1. Construct 15°

ANSWER Bisect 30°

2. Construct 75°

ANSWER $60^\circ + 15^\circ$

3. Construct an isosceles triangle with base 5 and lateral side 7

ANSWER Two arcs of 7

4. Why must the compasses open beyond half of AB ?

ANSWER Otherwise the arcs do not meet

5. Construct the centre of a given circle

ANSWER Two chords and their perpendicular bisectors

6. Construct a triangle from two sides and the included angle

ANSWER Copy the angle, step off the sides

7. Which construction cannot be done with these tools?

ANSWER Trisecting a general angle

07 Homework

Homework — 5 tasks

Leave every arc on the page; the arcs are the answer.

1. Construct the perpendicular bisector of a segment of 8 cm .
2. Construct the bisector of an angle of 80° .
3. Construct a triangle with sides 5 cm , 6 cm and 7 cm .
4. Construct an angle of 30° using only compasses and a straightedge.
5. Explain why no triangle can be constructed with sides 2, 3, 8.